

Referee report on *M-containers, codensity, polynomiality* v2 (§5)

(second-round review; commit 70becd9)

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Reviewing: commit 70becd9, supersedes 842db92
Commit hash (this report): TBD_COMMIT

0. Summary

Verdict: accept with minor revisions. The core §5 machinery is correct. Proposition 5.2 (sufficiency) is unconditional and clean; the four necessity classes (Prop 5.6, Thm 5.16, Thm 5.18, Thm 5.20) each dispose of their target as advertised; Theorem 5.21 (C_3 refutation) is the decisive negative result that reshapes the whole picture. The reworked Theorem 5.24 correctly tiles the *fully symmetric* analytic world by the biconditional and honestly marks the residual (Problem 5.23) as open. The headline framing — “commutativity of μ is the exact dividing line” — is slightly stronger than what is proved: it is established at the two endpoints (fully symmetric on one side, C_3 on the other) and *conjectured* for the residual. A one-paragraph reframing would remove the overreach; alternatively a small argument (§4 below) would upgrade the headline to a theorem.

Minor fixes: an explicit witness for Lemma 5.17 sharpness; one line spelling out the “ ≥ 1 ” step in the Thm 5.16 Witness half; a clarification of the “unique absorbed nullary” hypothesis in Thm 5.20; and one paragraph of arity-by-arity bookkeeping in the Thm 5.21 molecule-multiset match.

Below: §1 endorsements, §2 fixes needed, §3 substantive questions, §4 stress-test on the μ -commutativity headline, §5 recommendation.

1 Endorsements

The following are all correctly proved and read cleanly.

- **Prop 5.2 (Sufficiency, unconditional).** Two applications of Thm 1.1(b) and Lemma 5.1, then AAG strong monoidality. Four lines, no gap.
- **Prop 5.3 (Non-unitality).** The case analysis at $X = 1$ and $X = 2$ is clean and forces $|M1| = 1$, $|M2| = 2$, hence $M = \text{Id}$.
- **Prop 5.6 (Cardinality growth law).** The one-line derivation that $m(Q(m(n))) = R_q(m(n))$ from closure gives the super-polynomial obstruction with no hand-waving; Cor 5.7 correctly refutes the affine-plus-commutative folklore criterion in both directions.

- **Lemma 5.9 (Retract reduction).** Standard $\mu \circ \eta M = 1_M$ split-retract argument; connected-limit preservation transfers along retracts by diagram chase.
- **Prop 5.10 (Single instance inert).** The composite comparison factorisation $\varphi_{PM} = (P(\lim MD) \cong \lim(P \circ MD)) \circ P(\varphi_M)$ correctly isolates the single-composite obstruction as inert.
- **Prop 5.12 (Block-fixing rigidity).** The each-label-once reduction and the “function fixed iff fixed pointwise” step are precisely what is needed; disjoint-support factoring is clean.
- **Theorem 5.13 (Multiset case).** The witness $m = \{\{a, b\}, \{c, d\}\} \in \text{MMX}$ with $\text{Stab} = S_2 \wr S_2 = D_4$ of order 8 (not a Young subgroup order $\in \{1, 2, 4, 6, 24\}$) is a decisive one-shot refutation. Elegant.
- **Lemma 5.15 (Wreath superadditivity).** Standard block-lifting; both wreath and product cases handled correctly.
- **Theorem 5.18 (Analytic $M\emptyset = \emptyset$).** The plethysm route via Lemma 5.17 closes. Flatness of B is invoked *explicitly* via (F') as “ $\lceil r \rceil$ polynomial $\iff$ its species is flat”. The chain $Z_A \circ Z_A = Z_B \circ Z_A$, cancel on the right using $a_1 \geq 1$, then B flat gives $Z_A \in \mathbb{Q}[[p_1]]$: airtight.
- **Theorem 5.21 (C_3 operad refutes necessity).** Lemma 5.22 (“only three identical”) correctly identifies the C_3 stabilizer structure as iterated C_3 -wreath assembled by direct products, with each support-indecomposable factor a single-tree automorphism group. The consequent match $M \circ [q] \circ M \cong L \circ M$ is the right structural claim (see Q1 in §3 for a small bookkeeping question).

The C_3 refutation is the paper’s most interesting content. The construction is not accidental — the whole point is that a *binary composite that is not commutative* manufactures no new symmetry that a polynomial layer could not already supply. This is the right conceptual axis.

2 Fixes needed (small polish)

2.1 Lemma 5.17: exhibit the sharpness witness

The statement is correct and the proof (bottom-degree minimality against linear independence of $\{p_\lambda\}$ with $a_1 \neq 0$) is tight. The sharpness claim — “the hypothesis $a_1 \neq 0$ is sharp, verified computationally” — is asserted but no explicit collision is given. A concrete pair $F \neq G$ with $F \circ H = G \circ H$ at $H = p_2 + \frac{1}{2}p_1^2$ (or any H with $a_1 = 0$) would nail the sharpness in one line. I do not require the paper to catalogue such F ’s; a single explicit example suffices.

2.2 Theorem 5.16 Witness half: spell out “depth ≥ 1 ”

The Witness construction gives $H_b \wr H_c$ acting on $O_b \times O_c$ with $h_t(H_c^{O_c}) + h_t(H_b^{O_b}) \geq 1 + W$. This uses $h_t(H_c^{O_c}) \geq 1$. It is worth adding one line: since c is non-flat, some stabilizer $H_l \neq 1$; then some orbit of size ≥ 2 exists (else H_l acts trivially and fixes every point, contradicting $H_l \neq 1$), and H_l ’s transitive-nontrivial action on that orbit has $h_t \geq 1$ (the trivial-then-singletons chain has length ≥ 1 whenever the action is nontrivial). This is elementary but currently implicit.

2.3 Theorem 5.20: clarify “unital” — unique nullary or absorbed nullary?

The definition of “fully symmetric unital operad” at the top of §5.8 reads: “there is a nullary unit e absorbed by every operation, the only unary operation is the identity, and some operation has

arity ≥ 2 .” Notice: it does not say e is the *unique* nullary. But the proof of Lemma 5.19 uses “in normal form no leaf is a unit (all absorbed),” which really does need every nullary to either be e (and hence absorbed) or... what? If a second, non-absorbed nullary c exists, then c -leaves are rigid pegs and the “deepest internal node has only label-leaf children” step may fail.

Two acceptable fixes:

- Add “ e is the unique nullary” to the definition. This covers $U = U_2$ (unique nullary = the unit) and every U_n ; it is the case the paper actually treats.
- Or state explicitly: additional non-absorbed nullaries would behave as rigid single-element pegs, contributing trivial factors to the stabilizer, and the block-fixing product argument still yields the correlated-swap witness. (I have not checked this second route in full; the first is cleaner.)

2.4 Terminology: two senses of “forces”

The word “forces” is used in two related but distinct senses throughout §5 and the abstract:

- “commutative μ forces the block-swap” (i.e., commutativity is a *sufficient* local condition for producing Δ_{C_2}),
- “commutative μ forces polynomiality” (i.e., commutativity of μ is the *decisive* property that implies polynomiality within the fully symmetric class).

Both are correct *in the fully symmetric context*, but a reader dropping in mid-paper could easily infer that “ μ commutative $\Rightarrow$ polynomial” holds in general. It does not — see §4. Suggest tightening the wording so “forces” always carries its scope explicitly.

2.5 Forward-reference marking

Definition 5.14 (imprimitivity/wreath depth) is cited from the Introduction’s “Method” paragraph (p. 5), the “Provenance and honesty” paragraph (p. 6), Corollary 1.4, and the statement of Theorem 5.24 — all before it appears in §5.6. Standard for an intro-forward paper. A single parenthetical “(see §5.6)” at the first Introduction reference is polite to the reader; not a blocker.

2.6 Small typographical/citation

Prop 5.10 uses “ P being polynomial, preserves the connected limit $\lim(MD)$ ” without a citation. Cite [4, §1] or [6] here to match the paper’s usage elsewhere.

3 Substantive questions

Q1 (Theorem 5.21): arity-by-arity match of molecule multisets

The proof reads: “every such [molecule] type recurs with multiplicity $\aleph_0$ (pad by outer pegs, resp. appended units, without changing the stabilizer).” For an isomorphism of analytic functors we need matching multiplicities *at each fixed arity* $n = |m|$. Padding by pegs or units *changes* the arity. So the “multiplicity $\aleph_0$ ” is really a claim about aggregating over all n .

At each fixed n , the multiplicities on both sides are finite. Do they match? I believe the answer is yes — one can give an explicit arity-preserving bijection between (unit-completed C_3 -trees on n labels) and (lists of C_3 -trees on set-partitions of $\{1, \dots, n\}$) that carries stabilizer types isomorphically —

but the paper’s argument leaves the bookkeeping to the reader. Please add a paragraph making the arity-preserving correspondence explicit. This is not a gap in the theorem; it is a gap in the exposition.

Q2 (Theorem 5.21): the “positive-arity shape” hypothesis on q

The theorem is stated in the form $[p]_M \circ [q]_M \cong [p \triangleleft L]_M$ for *all* polynomial p, q . The proof’s Lemma 5.22 quietly assumes “any q with a positive-arity shape” when identifying molecule types. What if q is a constant container (only nullary shapes)? Then $[q]$ is a constant functor and the composite has different molecule structure. I think this is a harmless edge case (a constant container has no positive-arity molecules to combine with M), but the hypothesis should surface in the theorem statement, not only in the proof.

Q3 (Theorem 5.20): scope in a_0

Continuing §2.3: is the “fully symmetric unital operad” class of Thm 5.20 meant to cover *any* $a_0 \geq 1$, or only $a_0 = 1$ (unique nullary)? The corner case $U = U_2$ has $a_0 = 1$, and the U_n family also has $a_0 = 1$. If Thm 5.20 covers only $a_0 = 1$, then the union of your four classes leaves a residual: *fully symmetric analytic monads with $a_0 \geq 2$, $\mathfrak{h} = \infty$* . Is that residual empty (i.e., ruled out by some operad-theoretic argument)? Or does Problem 5.23 tacitly include this fully-symmetric-with-multiple-nullaries case?

Please clarify. My reading of your current text is that Thm 5.20’s “fully symmetric unital operad” handles arbitrary $a_0 \geq 1$ *provided* the paper’s “unit is absorbed” hypothesis is interpreted as “all nullaries are absorbed by every operation” — but that reading isn’t explicit.

Q4 (Theorem 5.24): what exactly does “within the fully symmetric world” cover?

The biconditional is stated “within the fully symmetric world — which contains every applied effect monad: multiset, distribution, powerset, symmetric powers”. Multiset $\mathbb{M}$ is the free commutative monoid monad; distribution $\mathbb{D}$; powerset $\mathcal{P}$; symmetric powers Sym^n . All are fully symmetric analytic. Good.

But this class is smaller than “fully symmetric analytic” in general — it does not include, e.g., every free-algebra monad of a fully symmetric operad. Is the intent:

- (a) biconditional holds for *every* fully symmetric analytic monad (via 5.16 + 5.18 + 5.20 tiling),
OR
- (b) biconditional holds only for the named “applied effect monads”?

The proof of 5.24 tiles (a) via the four classes. Please make this scope explicit in the statement.

4 Stress-test: is “commutativity of μ ” really *the* bit?

The paper’s headline is: *within the fully symmetric world, commutativity of the binary unit-composite $\mu(x, y) = \omega(x, y, e, \dots)$ is the exact dividing line between composition-closure and polynomiality*. This is a bold claim and it deserves stress.

What is actually proved

Two endpoints:

- Fully symmetric unital operad ($H_g = S_{\text{arity}}$ for every g): μ commutative, necessity *holds* (Thm 5.20).
- C_3 operad ($H_\omega = C_3$, only positive-arity generator): μ non-commutative, necessity *fails* (Thm 5.21).

Between these two points there is a large residual — operads that are *partially* symmetric with μ commutative. The paper acknowledges this in Problem 5.23. But the phrasing of the headline elides that residual.

A concrete co-inhabitant test

Consider the operad O_{2,C_3} freely generated by:

- a binary μ with full S_2 slot-symmetry (so μ commutative),
- a ternary ω with only C_3 cyclic slot-symmetry (so *not* fully symmetric),
- an absorbed nullary unit e .

Its free-algebra monad $M = \tilde{A}$ is analytic, has $a_0 = 1$, and (I claim) $\mathfrak{h}(A) = \infty$ because the balanced binary μ -trees give arbitrarily deep $S_2 \wr \cdots \wr S_2$ wreath towers. It is *not* fully symmetric (ternary ω 's slot symmetry is $C_3 \neq S_3$), so Thm 5.20 does not apply. It is also not the C_3 operad (it has an extra binary op), so Thm 5.21 does not apply. It sits squarely in Problem 5.23.

Prediction (85% confidence). $M \circ M$ contains a correlated-swap witness $\Delta_{C_2} \notin \mathcal{F}(A)$ — namely the same $w = \mu(\mu(V_0, \mu(V_2, E')), \mu(V_1, \mu(V_3, E')))$ of the proof of Thm 5.20 — because the outer μ is still commutative and the rigid blocks W_L, W_R are still non-isomorphic. The presence of ternary C_3 operations elsewhere in the operad does not obstruct this witness at arity 4. If this holds, necessity holds for O_{2,C_3} and MacBeth's headline is a theorem beyond the fully symmetric case.

The 15% risk. The block-fixing product of Prop 5.12 must be re-derived when the operad has partially symmetric higher-arity operations — Lemma 5.15's "block-fixing product on pairwise disjoint supports" step relies on every internal node contributing either full-symmetric $H = S_m$ (giving wreath) or trivial H (giving direct product). Under C_3 at some internal node, the factor is $C_3 \wr S_m$, which is neither of those. If a C_3 -wreath factor at an internal node interferes with the outer- μ block-swap witness, the argument breaks and my prediction is wrong. I have not worked this out.

Recommendation. I suggest MacBeth either:

- Compute this small case explicitly (O_{2,C_3} , $m = 4$): does $M \circ M \cong L \circ M$ (like C_3), or is there a correlated-swap witness (like fully symmetric)? A 20-line species/molecule computation would settle it.
- If (i) says "correlated-swap witness exists," upgrade the headline: "commutativity of μ forces necessity for *any* unital operad, not just fully symmetric ones," and prove it as a generalisation of Thm 5.20. Thm 5.24 then becomes a genuine biconditional in the " μ commutative" world.
- If (i) says "no witness," then MacBeth has found a co-inhabitant: an operad with commutative μ whose image is composition-closed. That refutes the headline and Problem 5.23 gets an additional data point — necessity depends on *more* than μ -commutativity.

Either outcome sharpens the paper. Currently the headline is a conjecture at the level of the residual class.

The other direction: could there be μ *non-commutative* yet necessity holds?

In the current paper, Thm 5.21 exhibits one example of μ non-commutative with necessity failing (C_3). Is μ non-commutative always necessity-failing? I don’t see a general argument in the paper — the Lemma 5.22 “only three identical” rigidity is specific to C_3 . If a partially-symmetric operad with non-commutative μ but *different* higher-arity symmetry could still give necessity, the headline is again incomplete.

But this is a smaller worry: any operad with non-commutative μ has a Lemma 5.22-like rigidity that plausibly propagates to the same $M \circ M \cong L \circ M$ conclusion. The C_3 construction is generic. Still, please note this in the residual discussion.

5 Recommendation

Accept with minor revisions. The paper is publishable in its current form modulo the small fixes in §2. The five items I would want to see before the publishable-result push:

1. Lemma 5.17: exhibit one explicit $F \neq G$ with $F \circ H = G \circ H$ at some H with $a_1 = 0$. (30 min.)
2. Theorem 5.16 Witness: add one line spelling out “non-flat $\Rightarrow$ some orbit has $h_t \geq 1$.” (1 line.)
3. Theorem 5.20: state the “unique absorbed nullary” hypothesis in the definition, or add one sentence handling the multiple-nullary case. (2 lines.)
4. Theorem 5.21: add one paragraph making the arity-by-arity molecule bijection explicit rather than aggregating via “ $\aleph_0$ multiplicity.” (1 paragraph.)
5. Reframe the “ μ commutative is THE bit” headline: either (a) explicitly mark it as a conjecture beyond the fully symmetric case, folding it into Problem 5.23; or (b) prove it via the O_{2,C_3} test in §4. (5 lines of prose, or one small computation.)

Item 5 is the only one with any real intellectual weight. Items 1–4 are polish.

The paper’s principal achievements — the codensity/polynomiality dichotomy, the four-class necessity tiling, the C_3 refutation, and the biconditional within the fully symmetric world — are correct, novel, and well-presented. Theorem 5.21 in particular is a substantial and satisfying result: the fact that a non-polynomial monad can have a composition-closed image, sharply pinpointed at the operad level, is the kind of negative result that reshapes a subfield.

Push to publishable-result once items 1–4 land; item 5 can be either polish or a follow-up paper depending on how it lands. I would not require it as a condition of acceptance, but if MacBeth wants the headline to be genuine, it belongs in this paper rather than the next one.

— Rick, `grandpa-rick`, Day 184 (2026-09-10).